

An Algebraic Specification of **fsb**

September 21, 2026

Abstract

This document specifies the security-relevant core of **fsb**, a local, read-only web view of a filesystem, as a small set of algebraic laws: the rule matcher that implements the two exclusion tiers (hide and deny), the access decision that every endpoint must agree with, the content-typed endpoints (images, PDFs, archives), the pure frontend functions (ordering, filtering, navigation, link resolution), and the preview state machine. Each law is tied to an executable check. Deriving the laws, and then auditing every place that compares names, exposed sixteen defects, eight of them bypasses or gaps in the access rules; they are recorded in [Section 9](#).

Contents

1	Purpose and Method	2
1.1	Why an algebra	2
1.2	Method	2
1.3	What is not covered	3
2	Sorts and Notation	3
2.1	Sorts	3
2.2	Environment	3
2.3	Operations	3
2.4	Conventions	3
3	Names and Folding	4
3.1	The name equivalence	4
4	The Rule Algebra	4
4.1	Definitions	4
4.2	Laws for a rule set without negation (Deny)	5
4.3	Laws for every rule set	5
4.4	Boundaries	6
5	The Access Decision	6
5.1	One question, asked once	6
5.2	Laws	7
5.3	Boundaries	9
6	Content-Typed Endpoints	9

7	Frontend Functions	10
7.1	Row order	10
7.2	Filter	10
7.3	Navigation	10
7.4	Markdown links and images	11
8	The Preview State Machine	11
8.1	States and events	11
8.2	Laws	12
9	Findings	12
9.1	Observations that are not defects	15
9.2	What the method did and did not find	16
10	Refinement and Summary	16
10.1	From the model to the code	16
10.2	How to run the checks	16
10.3	Summary	17

1 Purpose and Method

1.1 Why an algebra

fsb’s central promise is negative: a path that the user has denied is never listed, searched, previewed, downloaded, or even confirmed to exist. That promise is a statement about *every* way of reaching a file, so it is threatened whenever two code paths answer the same question differently. The defects found and fixed in the project so far had exactly that shape: a listing through a symlinked directory that leaked denied names, a trailing newline that defeated a rule, an unreadable file answering 403 instead of 404, and a macOS firmlink alias that bypassed a rule.

An algebraic specification forces the question to be written down once, and then every operation to be compared with it. This document does that for four parts of the system:

1. the rule matcher (Section 4);
2. the access decision shared by all endpoints (Section 5);
3. the content-typed endpoints and the pure frontend functions (Sections 6 and 7); and
4. the preview state machine (Section 8).

1.2 Method

Each law is stated in the form *for all inputs of a stated kind, a stated equation or implication holds*, then given an executable check: a randomized property test with a fixed seed, a test against a fixture that assembles the awkward cases together, or an end-to-end browser test. Laws that failed when first checked are recorded in Section 9 together with the fix. Laws that hold only under a stated assumption say so, and the assumptions that cannot be discharged inside the program are collected as *boundaries*.

1.3 What is not covered

The document does not model layout, streaming, timing, or the browser’s own behavior (except where the program relies on it). It does not replace an independent review; it is intended to give a reviewer a precise statement of what the program claims.

2 Sorts and Notation

2.1 Sorts

Name A path component: a finite sequence of Unicode characters containing neither `/` nor `NUL`.

Path An absolute path, written as the sequence of its components $p = \langle n_1, \dots, n_k \rangle$; the root is $\langle \rangle$. $\text{comps}(p)$ is that sequence, $\text{parent}(p) = \langle n_1, \dots, n_{k-1} \rangle$, and $p \cdot n$ appends a component.

Type $\{\text{dir}, \text{file}\}$, the type of the final component.

Rule A parsed pattern in the gitignore dialect, possibly negated and possibly restricted to directories.

RuleSet A finite sequence of rules. Two sets are used: **Deny** (hard block, no negation) and **Hide** (cosmetic, negation allowed).

Entry A directory entry as shown in a listing.

Row An entry as the frontend sorts it: name, type, size, and modification time.

2.2 Environment

Let **Roots** be the finite set of served roots, each already resolved to its real location. The filesystem is treated as a partial function from paths to files and directories with symbolic links; $\text{real}(p)$ is the canonical real location of p after all symbolic links and aliases (such as the macOS firmlink prefix `/System/Volumes/Data`) are resolved, defined when p exists. **canon** reduces aliases of the same location to one spelling.

2.3 Operations

The operations analyzed are, on the server:

Open, List, Search, and the endpoints built on them,

namely file download, head, metadata, image and PDF preview, and archive listing; and on the client the pure functions **cmp** and **sort** (row order), **filter**, **toHash** and **fromHash**, **goto**, and **resolveLink**.

2.4 Conventions

A relation $x \approx y$ between names means *the filesystem treats x and y as the same name*; it is defined in Section 3. A statement “**Open**(p) fails as **NotFound**” means that the operation fails with an error that the HTTP layer answers with an identical 404.

3 Names and Folding

3.1 The name equivalence

The rules are matched against names that the operating system will treat as identical. On a case-insensitive APFS volume two names denote the same file exactly when they are equal after Unicode normalization and *full* case folding. Full folding differs from simple folding: it expands some characters into several.

Definition 3.1 (Fold). $\text{fold}(s) = \text{nfc}(\text{cf}(\text{nfc}(s)))$, where cf is Unicode full case folding. For names, $x \approx y \iff \text{fold}(x) = \text{fold}(y)$. The extension to paths applies fold to the whole string, which is equivalent to folding component by component because folding never introduces or removes a separator.

Law 3.2 (Idempotence). $\text{fold}(\text{fold}(s)) = \text{fold}(s)$.

Observation 3.3 (Measured aliases). The set of single characters that APFS folds onto ASCII names was measured by creating every ASCII name of length one and two, and then probing every character in the Unicode range U+0080–U+2FFFF. On the tested system the aliases are exactly the following; each row names the ASCII spelling that the character opens.

character	opens
ß (sharp s), capital sharp s (U+1E9E)	ss
long s (U+017F)	s
Kelvin sign (U+212A)	k
ligatures U+FB00–U+FB06	ff, fi, fl, ffi, ffl, st

This is full case folding, not simple folding: under simple folding ß would remain a distinct character.

Law 3.4 (Fold agrees on the measured aliases). *For every pair in Observation 3.3, $\text{fold}(\text{character}) = \text{fold}(\text{ASCII spelling})$. The server's fold (Go, x/text/cases) and the client's fold (JavaScript, lower-, upper-, then lower-casing) both satisfy the law.*

Checked by: `rules/fold_test.go::TestFoldMatchesTheAliasesAPFS treats as equal; web/src/lib/laws.test.ts(filter agreement)`

Boundary 3.5 (Folding is a property of the volume). The alias set is a fact about the file system, not about fsb. A case-sensitive volume distinguishes more names and a future OS release could change the table. The specification therefore treats $\approx$ as the *coarsest* equivalence the program must respect: comparing after folding can only over-match, and over-matching a deny rule fails closed.

4 The Rule Algebra

4.1 Definitions

A rule r has a compiled test $\text{hit}(r, q, t)$: whether r selects the path q whose final component has type t . It is evaluated on folded text (Definition 3.1) with a pattern that was folded the same way.

Definition 4.1 (Exclusion). For a rule set $\mathcal{R}$ and a path p of type t , let $\text{anc}(p) = \{q \mid q \text{ is a prefix of } p\}$, where each proper prefix has type dir and p has type t . The last rule of $\mathcal{R}$ that hits q decides q : it excludes q unless it is negated. Then

$$\text{Match}(\mathcal{R}, p, t) = \exists q \in \text{anc}(p) : q \text{ is excluded by } \mathcal{R}.$$

Because an excluded ancestor already makes the whole path excluded, a later negation cannot re-include something inside it, as in git.

Definition 4.2 (Match below). $\text{MatchBelow}(\mathcal{R}, f, p, t)$ is Match restricted to the prefixes of p that are strictly longer than f . It is used to apply hide rules while listing a directory: entering a hidden directory on purpose shows its contents.

4.2 Laws for a rule set without negation (Deny)

Law 4.3 (Deny monotonicity). *If $\mathcal{R} \subseteq \mathcal{R}'$ then $\text{Match}(\mathcal{R}, p, t) \Rightarrow \text{Match}(\mathcal{R}', p, t)$. Adding a deny rule never makes anything accessible.*

Checked by: `rules/laws_test.go::TestLawDenyIsMonotoneAndOrderIndependent`

Law 4.4 (Order independence). *If $\mathcal{R}'$ is a permutation of $\mathcal{R}$, then $\text{Match}(\mathcal{R}, p, t) = \text{Match}(\mathcal{R}', p, t)$.*

Checked by: `rules/laws_test.go::TestLawDenyIsMonotoneAndOrderIndependent`

Law 4.5 (No negation in deny). *Parsing a deny file that contains a negated rule fails. Together with Laws 4.3 and 4.4 this gives the security reading of a deny file: it is a plain set of exclusions.*

Checked by: `rules/laws_test.go::TestLawDenyRefusesNegation`

4.3 Laws for every rule set

Law 4.6 (Ancestor closure). *If $\text{Match}(\mathcal{R}, p, \text{dir})$ then $\text{Match}(\mathcal{R}, p \cdot n, t)$ for every name n and type t . This holds also when $\mathcal{R}$ contains negated rules.*

Checked by: `rules/laws_test.go::TestLawAncestorClosure`

Law 4.7 (Fold invariance). *If $\text{fold}(p) = \text{fold}(p')$ then $\text{Match}(\mathcal{R}, p, t) = \text{Match}(\mathcal{R}, p', t)$: a path is excluded exactly when any spelling that the file system treats as the same name is. In particular no spelling of a denied name (upper case, decomposed accents, long s, Kelvin sign, sharp s, ligatures) escapes a rule.*

Checked by: `rules/laws_test.go::`

`TestLawMatchDependsOnlyOnTheFoldedPath;guard/foldalias_test.go`

Law 4.8 (Narrowing). $\text{MatchBelow}(\mathcal{R}, f, p, t) \Rightarrow \text{Match}(\mathcal{R}, p, t)$. *Ignoring ancestors can only remove matches.*

Checked by: `rules/laws_test.go::TestLawMatchBelowNeverAddsMatches`

Law 4.9 (Single component). *If a pattern contains neither $/$ nor $**$, then $\text{hit}(r, q, t)$ depends only on the final component of q . Glob classes never match $/$: a negated class excludes it, and a $/$ inside a class is a pattern error (fail closed).*

Checked by: `rules/laws_test.go::TestLawSlashlessPatternsStayWithinOneComponent;TestSlashInsideAClassIsAParseError`

Law 4.10 (Credential folders match wherever they are). *The core rules for credential folders (`.ssh`, `.aws`, `.gnupg`, `.kube`, `.config/gh`, Keychains, and the browser profiles) and files (`.netrc`) are not anchored to the current home directory: a path that contains one of them at any depth is excluded. The same secrets exist in a backup or clone of the home folder, in another account's home, and under a wrong `$HOME`, where an anchored rule protects nothing.*

Checked by:

`guard/differential\_test.go : TestCredentialFoldersAreDeniedWhereverTheyAre`

Law 4.11 (A rule file works as written or fails loudly). *For every rule file, either parsing fails with a message, or each rule line means what a reader would take it to mean. In particular no accepted line is a rule that can never match: a leading byte order mark and any line ending are normalized, and an inline comment, `~user/`, a quoted pattern, an invisible character or a lookalike of the slash are refused.*

Checked by: `rules/parse\_audit\_test.go`

Law 4.12 (The quick test never changes an answer). *Matching first asks whether any rule could match, using one combined expression per kind of rule, and tests slash-free patterns against the last path component only. These are optimizations: for every rule set without negation and every path, the result is that of consulting the rules one by one against the whole path.*

Checked by: `rules/laws\_test.go : TestLawFastPathEqualsTheRulesOneByOne`

Law 4.13 (Trailing whitespace is not significant). *For every rule r and path q , $\text{hit}(r, q \cdot \text{"n"}, t) = \text{hit}(r, q \cdot \text{"n "}, t)$ for names whose trailing characters are white space (space, tab, newline). macOS permits such names and they must not slip past a rule for `*.pem`.*

Checked by: `rules/rules\_test.go; guard/review\_test.go`

4.4 Boundaries

Boundary 4.14 (Hard links). Rules name paths, so a hard link to a denied file is invisible to them. The guard therefore judges a regular file with more than one name by its *identity* (device and inode): if any name of the file is denied, every name is refused, in `Open` and in listings alike (Law 5.5). The other names come from an index of all multi-name files under the roots and under the protected credential locations, built in the background because visiting every file takes seconds; until it is ready, and for a multi-name file that it has not seen yet (a link made since), the file is refused. Not detected: a denied name outside every root and protected location, and links missed because a walk was stopped by its limit.

Boundary 4.15 (Case classes in ranges). A positive class range such as `[+-0]` can include `/` by range even though the class text does not. No shipped rule uses such a range; a future tightening would expand ranges before compiling.

5 The Access Decision

5.1 One question, asked once

Every endpoint asks the same question of the guard: may this path be read? The answer is defined here and every operation is then compared with it.

Definition 5.1 (Access). Let $c = \text{canon}(\text{clean}(p))$ be the request path made absolute, cleaned, and reduced to one alias spelling, and let $\rho = \text{real}(c)$ be the real location of what is actually opened. The type t is the type of the opened object. Then

$$\text{Ok}(p) \iff \text{inRoots}(\rho) \wedge \neg \text{Match}(\text{Deny}, \rho, t) \wedge \neg \text{Match}(\text{Deny}, c, t).$$

The check is made on the *real* location (so a symbolic link cannot lead into a denied place), and also on the *spelled* location (so a link *named* like a denied file is refused even though its target is fine). It is therefore conservative: a path is refused if either name is denied.

Definition 5.2 (Open, list, search). $\text{Open}(p)$ returns the opened object, or `NotFound`, exactly when $\text{Ok}(p)$ fails or the path does not exist, and otherwise a permission or type error that mentions nothing about denied paths. $\text{List}(d)$ is defined when $\text{Open}(d)$ is; it contains an entry e of d exactly when $\text{Visible}(d, e)$:

$$\begin{aligned} \text{Visible}(d, e) \iff & \neg \text{Match}(\text{Deny}, \rho_d \cdot e, t) \wedge \neg \text{MatchBelow}(\text{Hide}, \rho_d, \rho_d \cdot e, t) \\ & \wedge (e \text{ a symbolic link} \Rightarrow \text{Ok-target}(e)), \end{aligned}$$

where $\rho_d = \text{real}(d)$: children are judged by where the directory really is, not by the link it was reached through. $\text{Search}(d, q)$ walks breadth-first from d , entering only visible unlinked directories, and reports the visible entries whose folded name contains $\text{fold}(q)$.

5.2 Laws

Law 5.3 (Denied is absent). *If $\neg \text{Ok}(p)$ because of a deny rule, then p is not opened, not listed by $\text{List}(\text{parent}(p))$, and not reported by any Search .*

Checked by: guard/laws_test.go :: TestLawDeniedIsAbsentEverywhere; guard/fuzz_test.go (independent inode and marker oracle)

Law 5.4 (Every spelling the OS resolves is refused). *For every secret s and every spelling p such that the operating system resolves p to the same object as s (case, folding aliases, decomposed accents, doubled separators, dot segments, trailing separators, the data-volume and symbolic-link aliases, named forks), every operation refuses p . The premise is decided by the OS (same inode), not by fsb, so the law does not depend on fsb's own idea of a name. Dually, every such spelling of an ordinary file inside the root is served.*

Checked by: guard/differential_test.go (both directions; fails 1,536 times if the folding fix is reverted)

Law 5.5 (Every name of a denied file is denied). *If a regular file has several names and one of them is matched by a deny rule, then Open refuses the file under every name, and no listing shows it. If the names cannot yet be determined, the file is refused. Identity is the pair (device, inode), never a hash of the contents, so an ordinary copy of a secret is not treated as the secret and no file is read to decide.*

Checked by: guard/hardlink_test.go (all four rule kinds, links made after the index was built, links to files outside the roots, refusal while the index builds, bounded walk); e2e 'a hard link to a denied file is refused under any name'

Law 5.6 (Indistinguishable from missing). *For every endpoint f , a denied path p and a missing path q receive the same status and body: $f(p) = f(q)$. This includes unreadable files inside a denied folder, which must not answer 403.*

Checked by: guard/laws_test.go :: TestLawDeniedIsAbsentEverywhere; server/m2_test.go; e2e 'answers 404, with no content, for denied paths on every endpoint'

Law 5.7 (Listed implies openable). *If e is listed in $\text{List}(d)$ and is not a dangling link, then $\text{Open}(d \cdot e)$ succeeds. (The converse is false by design: hidden entries are not listed but can be opened.)*

Checked by: `guard/laws_test.go::TestLawListedImpliesOpenable`

Law 5.8 (Found implies listed). If `Search` reports m , then m is an entry of `List(parent(m))`. `Search` therefore cannot show what browsing would not.

Checked by: `guard/laws_test.go::TestLawSearchResultsAreListed`

Law 5.9 (Hidden is reachable). If `Match(Hide, p, t)` but `Ok(p)` holds, then `Open(p)` succeeds. `Hide` rules are cosmetic and never restrict direct access.

Checked by: `guard/guard_test.go(hiddenEntries);e2e`lists the home folder without denied or ignored entries``

Law 5.10 (Containment). $\text{Ok}(p) \Rightarrow \exists r \in \text{Roots} : \text{real}(p)$ is r or lies beneath r , comparing folded text (so root membership respects the same equivalence as the rules).

Checked by: `guard/guard_test.go; guard/firmlink_darwin_test.go`

Law 5.11 (Containment is by identity). `inRoots( $\rho$ )` holds iff the leading components of ρ that correspond to a root name the same directory as that root (same device and inode). Comparing spellings would be right on a case-insensitive volume, but on a case-sensitive one it would admit a sibling folder that differs from the root only in case. This is an allow decision, so unlike the deny rules it must not over-match.

Checked by: `guard/casesens_test.go(needFSB_CASE_SENSITIVE_VOLUME); guard/guard_test.go::TestInRoots`

Law 5.12 (Alias reduction is by name equality). `canon` recognizes the data-volume prefix and the `firmlink` directories component by component, comparing components with `fold`, so no spelling of `/System/Volumes/Data/Users` escapes reduction. Slicing by byte length does not have this property, because a folded spelling can be longer or shorter.

Checked by: `guard/foldalias_test.go::TestRootOfSlashHonoursDenyRulesThroughFoldedFirmlinkSpellings; guard/firmlink_darwin_test.go`

Law 5.13 (The session is scoped to a random path). Browsers scope a cookie by host and never by port, so a cookie for `127.0.0.1` would accompany every request to every other web server on that address. All URLs of `fsb` therefore live under a random 128-bit prefix `/π/` fixed at launch; the cookie is set with `Path=/π/`; and a request whose path is not under the prefix is refused exactly like an unauthenticated one, whatever cookie it carries. A cookie that is sent only for paths under `/π/` is never sent to another server, which has no such path. The frontend forms every request relative to the page's own directory.

Checked by: `httpguard/httpguard_test.go::TestOnlyURLsUnderThePrefixExists, TestSessionCookieIsScopedToThePrefix; e2e`the session cookie is scoped to fsb's own prefix` (fail if the path is widened); web/src/lib/laws.test.ts::apiBase`

Law 5.14 (Canonical form is stable). $\text{canon}(\text{canon}(p)) = \text{canon}(p)$.

Checked by: `guard/laws_test.go::TestLawCanonPathIsIdempotent`

Law 5.15 (Bounded work). `Search` visits at most V directory entries and reports at most M results, and reports truncation when a bound stopped it; it never follows symbolic links to directories, so it terminates on cyclic links.

Checked by: `guard/search_test.go; e2e(100,000-filed directory, optional)`

5.3 Boundaries

Boundary 5.16 (Time of check). The file is opened first and its real path is then resolved and confirmed to name the same object (`SameFile`); the decision is taken on the object that is actually served. A rename between opening and resolving is treated as a race and refused.

Boundary 5.17 (Response timing). A denied path can be answered slightly faster than a missing one. Status and body are identical; timing is not equalized.

Boundary 5.18 (Other processes). Anything that runs as the user can read the same files. `fsb` is not a sandbox against local malware.

6 Content-Typed Endpoints

Three endpoints look inside a file: inline image preview, inline PDF, and archive listing. Each is defined by a predicate on the file's *bytes*, never its name, and each factors through `Open`, so Laws 5.3 and 5.6 apply to them unchanged.

Definition 6.1 (Sniff). `sniffimg` accepts a byte sequence that begins with the PNG, JPEG, GIF, or WebP signature; `sniffpdf` accepts one that begins with `%PDF-`; `sniffarc` accepts the start of a zip, a tar (`ustar` at offset 257), or a gzip stream that contains a tar.

Law 6.2 (Content decides). *The endpoint succeeds only if the corresponding sniff accepts the file's leading bytes; the file name is never consulted. A file named `a.png`, `a.pdf` or `a.zip` with other contents is refused with 415, and a file with a different name but the right contents is accepted. SVG and HTML are never accepted.*

Checked by: `guard/m2_test.go (OpenImage); guard/pdf_test.go; guard/archive_test.go; server/pdf_test.go`

Law 6.3 (Cloud-only files are never read). *For a file whose contents are stored only in the cloud, no endpoint reads it implicitly: the guard reports it as not downloaded before any read.*

Checked by: `guard/dataless_tests; e2e(hoverpeek)`

Law 6.4 (Archive listing discloses names only). *The archive listing contains, for each member, its name, size, type and date, and never any member data; names are displayed as text with control characters and invalid UTF-8 replaced, and are never used as paths.*

Checked by: `guard/archive_test.go:: TestListArchiveNamesAreDefanged; server/archive_test.go`

Law 6.5 (Archive listing is bounded). *For any file, listing scans at most E tar entries and streams at most D decompressed bytes, refuses a zip whose directory claims more than E entries, returns at most S entries, and reports when a limit stopped it.*

Checked by: `guard/archive_test.go:: TestListArchiveIsBounded, TestListArchiveStopAtTheDecompressionLimit`

Law 6.6 (Inline PDF is framed only by the app). *The PDF response carries `X-Frame-Options: SAMEORIGIN` and `frame-ancestors 'self'`, the app shell carries `DENY` and `frame-ancestors 'none'`, and the Markdown frame document carries its own policy that admits exactly its one script and one style by hash.*

Checked by: server/pdf_test.go; server/mdframe_test.go; web/mdframe_test.go; e2ePDF and Markdown tests

Boundary 6.7 (PDF sandboxing). A CSP `sandbox` cannot be applied to a PDF, because browsers then refuse to display it. The protection is instead the byte check, the framing headers, and the browser’s own isolated PDF viewer.

7 Frontend Functions

These functions are pure, so their laws are checked exhaustively enough by seeded random testing.

7.1 Row order

Let $\text{cmp}_{k,a,g}$ be the row comparison for sort key $k \in \{\text{name, size, time, kind}\}$, direction a , and grouping flag g (“Folders first”). A folder has no size and sorts as smaller than any file; a modification time that cannot be read sorts as the oldest; ties fall back to the name, then to the exact name.

Law 7.1 (Total preorder). *For every k, a, g , cmp is antisymmetric ($\text{cmp}(x, y) = -\text{cmp}(y, x)$), transitive, and total. This holds even for rows whose time is unparseable.*

Checked by: web/src/lib/laws_test.ts: compareRows is antisymmetric and transitive

Law 7.2 (Input order does not matter). *For distinct names, $\text{sort}(\pi(xs)) = \text{sort}(xs)$ for every permutation π . (Two names that differ only by normalization cannot coexist in one directory on APFS, so the premise excludes them.)*

Checked by: web/src/lib/laws_test.ts: sortRows is independent of input order

Law 7.3 (Grouping). *If g , every folder precedes every file, for every key and both directions; grouping does not flip with the direction.*

Checked by: web/src/lib/laws_test.ts: Folders first groups folders

Law 7.4 (Reversal). *If $\neg g$, the descending order is the reverse of the ascending order.*

Checked by: web/src/lib/laws_test.ts: descending is the reverse of ascending

7.2 Filter

Law 7.5 (Filter agrees with search). *For case-insensitive matching, $\text{filter}(n, q) \iff \text{fold}(n) \ni \text{fold}(q)$, where the client’s fold equals the server’s on the aliases of Observation 3.3. The filter box therefore finds the same names that filename search finds. With “Match case” both sides compare after normalization only.*

Checked by:

web/src/lib/laws_test.ts: the filter agrees with search; guard/matchcase_test.go

7.3 Navigation

Law 7.6 (Hash round trip). *$\text{fromHash}(\text{toHash}(p, s)) = (p, s)$ for every path and selection made of names, including names containing `%`, `#`, `?`, `&`, `=`, quotes, spaces, and non-ASCII characters. A literal `?` in a path is always encoded, so the first `?` of a hash starts the options.*

Checked by: `web/src/lib/laws.test.ts:parseHashInvertsPathToHash;format.test.ts`

Law 7.7 (Go to path is clean and idempotent). If $\text{goto}(s) = p$ then p is absolute, contains no `.`, `..` or empty component (except the root itself), never climbs above the root, and $\text{goto}(p) = p$.

Checked by:

`web/src/lib/laws.test.ts:resolveGotoIsCleanAndIdempotent;goto.test.ts`

Law 7.8 (Roots are prefix closed). p is within a root r iff $p = r$ or p begins with r followed by `/`, so a sibling sharing a prefix (`/Users/me` against `/Users/me`) is outside. The client check is advisory; the server's `inRoots` decides.

Checked by: `web/src/lib/laws.test.ts:withinRoots`

Law 7.9 (A name is shown as what it is). No character that reorders or hides text (controls, bidirectional overrides and isolates, line and paragraph separators, the byte order mark, tag characters) is sent to the screen inside a file name: each is replaced by a visible marker naming its code point. Text that needs none of them, including right-to-left scripts, is unchanged, and links and downloads use the real name.

Checked by: `web/src/lib/laws.test.ts:`

`displayName;e2e`names that would display themselves are shown with visible markers``

7.4 Markdown links and images

Law 7.10 (Links resolve to normalized absolute paths). If `resolveLink(b, h)` names a file, the path is absolute and contains no `.`, `..` or empty component; relative targets are resolved against the directory b of the Markdown file and cannot climb above the root.

Checked by: `web/src/lib/laws.test.ts:`

`resolveLinkYieldsNormalizedAbsolutePath;markdown.test.ts`

Law 7.11 (Links and images carry no address). The generated markup has no `href` and no `src` on links or images before sanitizing; remote images are replaced by a placeholder, raw HTML is escaped, and only `http` and `https` links are ever opened, in a new tab without a referrer. Only base64 raster `data:` URLs fetched through the guarded preview endpoint are put into images.

Checked by: `web/src/lib/markdown.test.ts;e2eMarkdownTests`

8 The Preview State Machine

8.1 States and events

For a selected entry the preview pane is in one of the states

idle, loading, folder, image, pdf, archive, text, binary, empty, dataless, error.

The name of the entry gives a *hint* $h \in \{\text{image, pdf, archive, none}\}$; it is only a hint because the server decides from the bytes. Selecting an entry moves to loading, after a short delay that keeps holding an arrow key from firing a request per row, and then:

- a folder goes to folder; a dangling link to error;

- with hint image, the pane shows `image` and, if the server refuses the bytes (415), falls back to the head request;
- with hint pdf, a one-byte range request asks the server; a 200/206 of type `application/pdf` gives `pdf`, a 409 gives `dataless`, anything else falls back to the head request;
- with hint archive, the listing request gives `archive`, a 409 gives `dataless`, anything else falls back to the head request;
- otherwise the head request gives one of `text`, `binary`, `empty`, `dataless`, or, on failure, `error`.

8.2 Laws

Law 8.1 (Termination). *From loading every run reaches a terminal state (`folder`, `image`, `pdf`, `archive`, `text`, `binary`, `empty`, `dataless`, or `error`) after at most two requests: one hinted request and at most one fall back to the head request, which never falls back again.*

Law 8.2 (Freshness). *The content shown always belongs to the entry that is currently selected: every request started for a selection, including the fall back after an image fails to load, is cancelled when the selection changes, and a cancelled request never changes the state.*

Law 8.3 (Frame only after confirmation). *An inline frame is created for a PDF only in state `pdf`, which is entered only after the server has answered `application/pdf` for that path; a Markdown frame is created only for a text head whose name is a Markdown name, and receives only sanitized HTML.*

Checked by: e2e: ‘‘shows area lPDF in the built-in viewer frame, and treats a fake. pdf as text’’; Markdown tests

Law 8.4 (Nothing is shown for a denied entry). *Every request of the state machine goes through the endpoints of Section 5; for a denied entry each answers as for a missing one (Law 5.6), so the pane shows only “Not available”.*

Checked by: e2e: never lists secret files; answers 404

Boundary 8.5 (Termination and freshness are argued, not enumerated). Laws 8.1 and 8.2 are properties of the component’s control flow. Freshness was violated by one path (Finding 1) and is now guaranteed by keeping a single cancellation handle per selection; there is no automated test for that race, because it needs a controllable slow response. It is exercised indirectly by the end-to-end tests that select files quickly.

9 Findings

Stating the laws and then checking them found six defects (1–6), and a subsequent audit of every place that compares names or paths, and of the environment assumptions behind the rules, found ten more (7–16). Each was reproduced by a failing test before it was fixed. Findings 1, 7, 8 and 10 to 15 are security defects; the rest are consistency defects that fail closed or are cosmetic, but each is a place where two parts of the program disagreed.

	Defect	Law broken	Fix
1	Deny bypass through full case folding. On APFS the name <code>.fh</code> opens <code>.ssh</code> , but the matcher folded with simple folding, so a rule for <code>~/ssh</code> did not match it; <code>Open</code> , <code>Head</code> and the listing of the parent all served the denied folder. Also long s, the Kelvin sign and the ligatures.	4.7, 5.3	Rules, roots and search compare after full folding (<code>rules.Fold</code>); the alias set was measured on this file system and is a test.
2	Listed but unopenable. A plain file named like a <code>dir/</code> deny rule (a file called <code>cache</code>) was shown by <code>List</code> but refused by <code>Open</code> , because <code>Open</code> 's pre-check assumed every path is a directory.	5.7	The pre-check treats the final component as a file; the precise check after opening still refuses real directories.
3	Glob class matched across a slash. <code>[!a]</code> compiled to <code>[^a]</code> , which matches <code>/</code> , so a pattern with no slash could match across directory components.	4.9	Negated classes exclude <code>/</code> ; a <code>/</code> inside a class is a pattern error.
4	Row order not transitive. A modified time that could not be parsed compared equal to everything, so <code>a=b</code> , <code>a=c</code> but <code>b<c</code> . The resulting order depended on the input order.	7.1, 7.2	Unreadable times sort as the oldest; comparisons no longer use subtraction.
5	Filter and search disagreed. The filter box folded with simple lower-casing, so <code>strasse</code> did not find <code>Straße</code> , while filename search did.	7.5	The filter folds with lower-, upper-, lower-casing, which matches full folding on the measured aliases.
6	Stale fallback. After an image failed to load, the fall-back request had its own cancellation handle, so a slow answer could replace the preview of a newer selection.	8.2	One handle per selection is shared by every request started for it.

	Defect	Law broken	Fix
7	Firmlink prefix matched by byte length. The data-volume prefix and the firmlink directories were recognized by slicing the path to the prefix's length, so a folded spelling (<code>/Slong-system</code> , or <code>Data</code> with the <code>st</code> ligature) was cut mid-character and not reduced. Under <code>-root /</code> the unreduced path was inside the root and matched no rule.	5.12, 5.3	Components are compared one at a time with <code>Fold</code> .
8	Root containment by spelling. Containment folded names, so on a case-sensitive volume a root <code>Public</code> also served the different folder <code>public</code> . Folding is safe for deny rules but not for an allow check.	5.11, 5.10	Containment is decided by identity: the corresponding leading directory must be the root itself.
9	Client and CLI comparisons. Breadcrumbs sliced the path by the root's character count (wrong when folding changes length) and compared roots case-sensitively; a link or hash naming a file with another case or accent selected nothing; the core-rule warning compared rule text exactly.	7.8	All use the same name equality; breadcrumbs split by components.
10	Credential rules anchored to one home directory. The rules for <code>~/.ssh</code> , <code>~/.aws</code> and the others named the current home only, so the same folders in a backup or clone of the home folder, in another account's home, or under a wrong <code>\$HOME</code> were served.	4.10	The credential rules match at any depth, like the secret-file patterns.
11	A path is not a file (hard links). Rules name paths, so a hard link to a credential file placed anywhere else was served under its innocent name; this had been recorded only as a limitation.	5.3, 5.5	A file with several names is judged by identity: a denied name denies them all, in <code>Open</code> and in listings. First done for the credential locations only.

	Defect	Law broken	Fix
12	The session cookie followed the host, not the port. Cookies are scoped by host, so the cookie was also sent to every other web server on 127.0.0.1 that the same browser visited, which could replay it. It had been recorded as a limitation.	5.13	Everything is served under a random per-launch prefix and the cookie's path is that prefix; requests outside it are refused.
13	Hard links to .env, .pem and .key files. The identity check of finding 11 covered only the credential folders, so a second name for a secret file elsewhere (<code>notes.txt</code> linked to a <code>.env</code>) was still served.	5.5	The check covers every denied name under the roots and protected locations, using a background index of multi-name files; multi-name files are refused until the index is ready.
14	Rule files that load and do nothing. A byte order mark broke the first rule; bare carriage returns joined lines; an inline comment, <code>~user/</code> , quotes, an invisible character or a lookalike slash produced a rule that never matched. All loaded without an error.	4.11	The byte order mark and line endings are normalized; the rest are errors that say what is wrong, so fsb does not start.
15	Secrets under other names. Only known folders and the extensions <code>.pem</code> and <code>.key</code> were denied, so a private key copied out of <code>.ssh</code> (<code>id_rsa</code>), a certificate store or a password database was served.	4.10	The names that identify such files are core rules at any depth (<code>.pub</code> files stay reachable).
16	A name that shows as another name. A right-to-left override or a control character in a file name reordered or hid what the listing displayed, so <code>invoiceU+202Etxt.exe</code> read as a text file.	7.9	Deceptive characters are replaced, for display only, by a visible marker; real right-to-left text is untouched and downloads keep the real name.

9.1 Observations that are not defects

- A dangling symbolic link is listed (marked as broken) but cannot be opened. Law 5.7 excludes it.

- Hidden entries are not listed but can be opened, and a hidden directory that is entered on purpose shows its contents. (Law 5.9.)
- A symbolic link *named* like a denied file is refused even when its target is harmless. Definition 5.1 checks both names.
- The macOS by-inode path `/.vol/<device>/<inode>` was probed as a possible alias that would evade path rules; on the tested system the directory is empty and the path cannot be opened.

9.2 What the method did and did not find

Findings 1 to 3 come from the rule and access laws and are about names: two spellings, two types, two segments treated as one. Findings 4 to 6 come from the frontend laws. Findings 7 to 9 came from asking of every comparison in the code whether it assumes that the same name means the same bytes, and from checking each answer against the real file system (including a case-sensitive volume). Findings 1, 7 and 8 share one lesson: name equality is a property of the volume, so deny checks may over-match but allow checks must decide by identity. The laws that held on first check (monotonicity, order independence, ancestor closure, narrowing, search implies listed, denied is absent, the round trips) are as informative as the failures, since they turn long-standing assumptions into checked statements.

10 Refinement and Summary

10.1 From the model to the code

Abstract	Concrete
fold	<code>rules.Fold (Go)</code> ; fold in <code>web/src/lib/filter.ts</code>
Match, MatchBelow	<code>Set.Match</code> , <code>Set.MatchBelow</code> in <code>internal/rules</code>
Deny, Hide	the two <code>rules.Set</code> values held by <code>guard.Guard</code>
canon, real	<code>canonPath</code> , <code>filepath.EvalSymlinks</code> in <code>internal/guard</code>
Ok, Open	<code>Guard.open</code> , with violation and <code>deniedByAncestor</code>
List, Visible	<code>Guard.ListFunc</code> , <code>Guard.entry</code>
Search	<code>Guard.Search</code> (uses <code>entry</code>)
sniff	<code>sniffImage</code> , <code>OpenPDF</code> , <code>ListArchive</code>
cmp, sort	<code>compareRows</code> , <code>sortRows</code> in <code>sort.ts</code>
filter	<code>matchesName</code> in <code>filter.ts</code>
toHash, fromHash	<code>pathToHash</code> , <code>parseHash</code> in <code>format.ts</code>
goto	<code>resolveGoto</code> in <code>goto.ts</code>
resolveLink	<code>resolveLink</code> in <code>markdown.ts</code>

Every HTTP-facing endpoint reaches the file system only through `Guard`; a test fails the build if an HTTP-facing package imports `os`, `io/fs`, `syscall` or `path/filepath`. This is what allows Laws 5.3 and 5.6 to be stated once for the guard rather than once per endpoint.

10.2 How to run the checks

```

go test -race ./...           # rule, guard and server laws
cd web && npm test             # frontend laws
node --test e2e/chrome.test.mjs # end-to-end (Safari: e2e/safari.test.mjs)
make -C specs/algebra         # build this document

```

10.3 Summary

- The deny rules form a set of exclusions: monotone, order independent, closed under descent, and invariant under every spelling that the file system equates.
- Every endpoint agrees with one access decision; listing implies openable, search implies listed, and denied paths are absent and look missing.
- Content-typed endpoints decide from bytes, disclose names only, and are bounded.
- The frontend order is a total preorder independent of input order, and the filter, hash and link functions are the exact counterparts of their server-side definitions.
- Sixteen defects were found and fixed; the laws that hold are recorded with the checks that establish them.

The document is intended to be read alongside `SECURITY.md` and the product functional and design specifications, which sit in the enclosing `specs` folder: those state what is promised and how it is built, and this states, precisely, the properties from which the promise follows.